**Result 1 (continuous identity fixes each point).** [Mathlib, ContinuousMap.id_apply]
Let $X$ be a topological space. Let $x$ be a point of $X$. Then

$$\mathrm{id}_X(x) = x.$$

Adapted from the cited source.

**Result 2 (continuous composition acts pointwise).** [Mathlib, ContinuousMap.comp_apply]
Let $X$, $Y$ and $Z$ be topological spaces. Let $f$ be a continuous map from $X$ to $Y$. Let $g$ be a continuous map from $Y$ to $Z$. Let $x$ be a point of $X$. Then

$$(g \circ f)(x) = g(f(x)).$$

Adapted from the cited source.

**Result 3 (identity on the target preserves a continuous map).** [Mathlib, ContinuousMap.id_comp]
Let $X$ and $Y$ be topological spaces. Let $f$ be a continuous map from $X$ to $Y$. Then

$$(\mathrm{id}_Y \circ f) = f.$$

Adapted from the cited source.

**Result 4 (identity diagram has its declared spaces).** [Topology foundations, identity diagram projections]
Let $X$ be a topological space. Let $D = \text{identity diagram}(X)$. Then

$$D.\text{source} = X \text{ and } D.\text{target} = X.$$

**Result 5 (diagram supplies its continuous map).** [Topology foundations, dependent diagram field]
Let $D$ be a diagram consisting of one continuous map. Then

$$\left(\text{there exists } f : C(D.\text{source}, D.\text{target}),\ f = D.\text{arrow}\right).$$

LLM-assisted Clean specimens and proofs (gpt-6, 2026-11-06). Mathematical sources are attributed above; see PROVENANCE.md.